Eli Young

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Education

Swarthmore College — Swarthmore, PA
Candidate for B.A. in Computer Science; Cumulative GPA: 3.12/4.0; Major GPA: 3.5/4.0 — *May 2026*

Relevant Coursework: ML, Algorithms, Compilers, DS, Systems, Big Data, Discrete Math, LinAlg, Calc I–III

Skills

Languages: Python, C++, JavaScript/TypeScript, Java, C, C#, SQL, R, Bash, Swift, OCaml
Technologies: React, Node.js, Flask, Docker, AWS, Git, Firebase, PyTorch, TensorFlow, scikit-learn

Relevant Experience

AlgoArena, LLC | **algoarena.net** — Remote
Co-Founder & Full-Stack Developer — *Sept. 2025 – Present*

- Co-founded real-time 1v1 competitive programming platform (5,000+ problems, ELO matchmaking); full-stack Next.js 14/TypeScript/Firebase, Firestore, Stripe/SendGrid, security rules.
- Integrated OneCompiler API (16 languages) and AI mock interviewer (Deepseek, Cartesia TTS) for 24/7 voice-based technical interview practice with streaming responses.

USDA Agricultural Research Service & Swarthmore College — Swarthmore, PA
Applied Machine Learning Researcher — *Sept. 2025 – Present*

- Engineered time-series features for EPG analysis (FFT, wavelets, sliding windows); Optuna/SMOTE pipelines achieved 31% F1 gain (0.43→0.57) via novel position-encoding features.
- Benchmarked MOMENT and AdaBoost; USDA ARS, Auburn, Harvey Mudd on interpretable crop health.

Calibrr Social — Remote
Contract Software Engineer — *June 2025 – Jan. 2026*

- Reduced API rate limiting with debouncing and cache optimization, cutting API calls by 75% (from 60/min to 10–15/min); enabled profile likes and smart email notifications via EC2, AWS Lambda, and SES.

Center for Digital Agriculture, University of Illinois Urbana-Champaign — Urbana-Champaign, IL
Research Assistant, REU Fellowship — *May 2025 – Aug. 2025*

- ML on 690 eggs' VNIR spectroscopy for chick mortality and sex prediction (transformers, meta-learning, TabPFN, gradient boosting); PCA (PC1 76.7%) for wavelength selection, addressing 6.5B+ annual culls.
- Achieved 98.8% mortality and 59.5% sex accuracy (100% on confident predictions); validation simulated hatchery ops with perfect viable-egg detection, demonstrating commercial viability.

Torchman I.P. — Remote
Software Engineering Intern — *Sept. 2024 – Apr. 2025*

- Pipeline for USPTO office actions: OCR (Tesseract), regex, OpenAI API, Selenium/BeautifulSoup; CSV export, summary generation, and PDF retrieval from Google Patents.

Projects

Lock In ***trylockin.com*** — May 2024 – Present
Cross-Platform Productivity App with Native App Blocking

- React Native productivity app with native Kotlin/Swift app blocking; Firebase, Google/Apple Sign-In, AdMob, in-app purchases; gamification (currency, streaks, D3.js dashboards).

Precise Swine & Dine ***GitHub*** — Jun. 2025
Computer Vision & ML for Precision Livestock Management

- YOLO + morphological features for pig weight estimation (<5 kg MAE); MLP/Random Forest on 9,579 images. Won Livestock Monitoring Track, AI Foundry for Ag Applications Hackathon.

Leadership & Additional Experience

ICPC Team Member *(5th place, ICPC NA South Division 2 — Mid-Atlantic Regional, Wilkes Univ.)* *Sept. 2024 — Nov. 2024*